Verdict Project Roadmap

Gridline — FIA penalty consistency tool | Personal reference — September 2026

The user has an explicit goal for this project to be ML/AI heavy overall. Each phase below is tagged with an honest AI/ML fit — where it's genuinely called for, where it's an optional well-justified addition, and where forcing it in would just be worse engineering.

Phase 1 — PDF Parsing

Status: Done

AI/ML: Optional AI angle — core parser is correctly rule-based (fixed template); a fallback LLM extractor for documents that don't match any known template would be a legitimate future addition, not a replacement.

Done:

- Extract text from FIA decision PDFs (pdfplumber)
- Split text into structured fields: Driver, Competitor, Session, Fact, Infringement, Decision, Reason
- Handle label wording variance across seasons (Infringement / Offence normalized to one field)
- Extract turn number, own car, other car, and penalty points via regex
- Fallback logic: pull "other car" from Reason when Fact doesn't mention a second car
- Batch-run the parser across a folder of PDFs, fault-tolerant per file
- Validated against four real, structurally different documents (grid drop + points, warning, no further action, older wording)
- Automated pytest coverage for all four validated documents

Remaining:

- Handle multi-driver "No / Driver" documents (rare, untested edge case)

Phase 2 — Storage

Status: Done

AI/ML: No AI fit — pure data engineering (JSON files, Postgres). Nothing here should be ML.

Done:

- Save each parsed decision as a JSON file (verdict/data/parsed/)
- Design and run a Postgres migration (db/sql/010_penalty_decisions.sql) following the existing db/sql/ pattern
- Loader script (verdict/loader/load_to_postgres.py) reads JSON into Postgres, safe to re-run without duplicates
- Verified end-to-end: 4 PDFs → 4 JSON files → 4 clean database rows

Remaining:

- Run the parser across a much larger batch of real documents (dozens-hundreds, multiple seasons) as more edge cases surface — schema can evolve cheaply since it's a solo project

Phase 3 — Telemetry

Status: Barely Started

AI/ML: Optional AI angle — mostly careful data engineering/math; if wall-clock-to-lap-time alignment is ever ambiguous, signal/pattern matching (braking or speed signature matching) could resolve it.

Done:

- Proof-of-concept script pulling FastF1 lap telemetry (speed, throttle, brake, position) for a known driver/lap

Remaining:

- Build the mapping from a parsed penalty record (race, driver, incident time) to the matching FastF1 session/lap
- Pull telemetry (speed, position, gap, who was ahead) for that exact moment
- This time-alignment step is a real unknown: FIA documents use wall-clock time, FastF1 telemetry is lap/session-relative

Phase 4 — Join

Status: Not Started

AI/ML: No AI fit — merging two already-structured datasets. Straight data engineering.

Remaining:

- Attach telemetry data to each penalty record in one combined structure or table
- Depends entirely on Phase 3's time-alignment work being solved first

Phase 5 — Comparison / Consistency Model

Status: Not Started

AI/ML: Core ML phase — mandatory, not optional — feature engineering from telemetry + incident data, similarity/clustering, likely a trained classifier comparing predictions against real verdicts. This is the project's real ML learning opportunity.

Remaining:

- Define what "similar incident" means (turn type, closing speed, contact severity, who was ahead, etc.)
- Build the actual logic that flags cases where similar incidents received different verdicts
- Most conceptually difficult phase of the project — an analysis/design problem, not just code

Phase 6 — Searchable UI

Status: Not Started

AI/ML: Optional AI angle — semantic search via text embeddings (so a search finds relevant incidents by meaning, not just exact keywords) would be a well-scoped NLP addition here.

Remaining:

- Frontend to search and browse incidents, and surface flagged inconsistencies
- Likely extends the existing Gridline React app rather than a new one

Phase 7 — Productionizing

Status: Not Started

AI/ML: No AI fit — Airflow/Docker/deployment. Infra work, not modeling.

Remaining:

- Wire the parser into an Airflow DAG (matching existing DAGs) to run on a schedule

- Deploy alongside the rest of Gridline via the existing Docker Compose setup
- Handle new FIA documents automatically as each race weekend publishes them

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Reality Check

This is a multi-week project, not a few more sessions. Phases 1 and 2 (parsing and storage) are done, but they were also the most mechanical phases. The three hardest, most uncertain phases — telemetry-to-verdict time alignment, defining incident "similarity," and the comparison model itself — haven't been touched yet, and each is its own multi-step effort with genuine unknowns, not just functions to write.

Recommended Next Checkpoint

Not "production" — a proof point. Pull matching FastF1 telemetry for 2–3 of the already-parsed decisions, and get one incident fully joined end-to-end (PDF verdict + telemetry side by side). This proves the biggest technical unknown (time alignment) before investing further in parsing scale or the comparison model.